

DATA MINING PROJECT
Master in Data Science and Advanced Analytics

NOVA Information Management School
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Clustering Analysis for Customer Segmentation at Amazing International Airlines Inc.

Group 44

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Índice

1. Executive Summary.....	3
2. Introduction.....	3
3. Methodology.....	3
3.1. Data Preparation and Preprocessing.....	3
3.2. Outlier Analysis and Treatment.....	4
3.3. Feature Selection and Engineering.....	4
3.5. Segmentation perspectives.....	4
3.4. Clustering algorithms.....	5
4. Results & Validation.....	5
4.1. K-means.....	5
4.2. Hierarchical Clustering.....	6
4.3. Self-Organizing maps.....	7
4.4 Merging Clusters analysis.....	9
5. Strategic Recommendations.....	9
6. Conclusion.....	10
7. Annexes:.....	11
7.1. Annex Statements.....	11
7.1.1. AI Usage Statement.....	11
7.1.2. Contribution Statement.....	11
7.1.3. Responsibility Statement.....	11
7.2 Annex: Dataset Information.....	11
7.3. Figures.....	14
7.4. Tables.....	24

1. Executive Summary

This report presents a data-driven customer segmentation for Amazing International Airlines Inc. (AIAI) to enhance personalized marketing and customer retention. Using a dataset of loyalty program members and flight activity over three years, we developed a multi-perspective clustering approach. Following a rigorous data preprocessing and feature engineering phase, we applied *K-Means*, *Hierarchical Clustering* and *Self-Organizing Maps (SOM)* across three distinct lenses: **Demographic** and **Behavioral** patterns.

Our analysis identified five primary customer segments: **Consistent Loyalties**, **High-Value potential**, **Solo Travelers** and **Low-Engagement users**. The segments were validated through internal metrics (*Silhouette Score* and R^2) and business interpretability. The results reveal critical differences in booking frequency, companion travel, and loyalty point redemption. Based on these insights, we propose a strategic roadmap focused on tiered rewards for high-value segments and re-engagement campaigns for at-risk customers. These recommendations aim to maximize AIAI's revenue potential and foster long-term loyalty in a competitive market.

2. Introduction

Aurora International Airlines works in a very competitive market, where understanding customers and keeping their loyalty are important for long-term success. As the airline industry changes, using data to analyze customer behavior has become an important way to improve relationships and find useful insights about different types of customers. Some travelers look for comfort and high-quality service, while others care more about price or how often they travel. This project lies the gap between static **demographic profiles** and **dynamic flight behaviors**. By integrating these disparate data sources and applying different types of clustering, we aim to identify distinct passenger segments based on spending, frequency, and seasonality. This analysis moves beyond intuition, providing AIAI with a rigorous, data-driven foundation to tailor marketing strategies. Ultimately, these findings will enable the airline to deliver personalized value, transforming raw data into a competitive advantage for retention and growth.

3. Methodology

3.1. Data Preparation and Preprocessing

The Data Preparation as an initial phase was critical to ensure the data quality and reliability for the next segmentation phase. For this process we began merging the two datasets: **CustomerDB** and **FlightsBD**. Since the flight data contained multiple entries per customer, an aggregation strategy was implemented to summarize flight behaviors (e.g., total flights per customer, distance per customer,...) and derive seasonal patterns (e.g., flights per season each customer) before merging with the customer profile data.

Some inconsistencies were found, so to address this problem some data cleaning was made such as:

- We identified that the feature **Loyalty#** had some duplicate entries, representing approximately 0.97% of the dataset. These were removed to ensure unique customer profiles.
- For missing values we noticed that the rows with missing values in key columns like **Income** were found to be empty across multiple attributes and were consequently dropped. For the missing

values in the feature **CancellationDate** we assumed that the customers never cancelled the loyalty program.

- Inconsistencies between **EnrollmentDateOpening** and **CancellationDate** were detected where the cancellation date was earlier than the actually starting in the loyalty program, this was corrected by swapping reversed dates or removing invalid entries.

3.2. Outlier Analysis and Treatment

For the numerical outliers, an initial analysis using boxplots revealed that outliers constituted roughly 23% of the dataset. To mitigate their impact without significant information loss, the *Capping (Winsorizing)* method was applied, limiting extreme values to the 1st and 99th percentiles.

While univariate analysis identifies extreme values in single features, we used *DBSCAN* to detect multivariate outliers. This algorithm is ideal because it identifies "noise" points that do not belong to any dense cluster in high-dimensional space. Unlike distance-based methods, *DBSCAN* does not assume spherical clusters and can find anomalies based on unusual combinations of features. We determined the optimal *Epsilon* using a *K-Distance* graph, identifying the "elbow" where point density drops significantly. Points labeled as -1 were classified as outliers, representing observations with unique behavioral patterns that could distort the final clustering. By isolating these 0.05% of cases, we ensured the resulting segments were representative of the broader customer base.

3.3. Feature Selection and Engineering

For feature engineering some new features were created to enrich the analysis, such as **ProgramDaysDuration** (number of days that the customer has joined the loyalty program) and **IsCanceled** (indicates whether a customer canceled or not his loyalty program).

Numerical features needed to be scaled to ensure that features with larger ranges did not dominate the distance calculation and provoke bias, for that we compared *MinMaxScaler* and *StandardScaler*. The *StandardScaler* (Z-Score scaling) was selected as it effectively centered the data distribution around zero, providing a balanced normalization for the clustering algorithms.

To conclude redundant or non-informative variables (e.g., **Country**, **City**, **Latitude** / **Longitude**, and identifiers like names) were removed. Highly correlated features, such as **TotalDollarCostPointsRedeemed** (perfectly correlated with **TotalPointsRedeemed**), were dropped to avoid multicollinearity.

3.5. Segmentation perspectives

Before starting our analysis it was necessary to create some segmentation perspectives which is the process of dividing a large group of customers into smaller, specific groups based on shared characteristics. This helps us understand patterns in the data and identify different types of customers. For this analysis, we are looking at customers from: a **Demographic perspective** and a **Behavioral perspective**. By combining these two views, we can see both who the customers are in terms of value and how they like to travel.

First, the **Demographic perspective** groups customers based on their economic value. This looks at their personal and financial standing to see how profitable they are to the company. The specific features we chose for this are **Income**, **Customer Lifetime Value**, and **ProgramDaysDuration**. We also included **TotalFlights** and **TotalPointsRedeemed** because these help measure the overall value and activity level of the customer.

Second, the **Behavioral perspective** groups customers based on their flight preferences and habits. Instead of just looking at how much money they spend, this looks at *when* and *how* they travel. To understand this, we selected variables like the number of flights taken in each season (**Flights_Summer**, **Flights_Spring**, **Flights_Winter**, and **Flights_Fall**). We also look at their travel style using **TotalFlightsWithCompanions** and **SoloFlightRatio** to see if they fly alone or with others. Finally, we track their engagement intensity with features like **PointsRatio**, **MaxFlightsInMonth**, and **MaxPointsInMonth**.

3.4. Clustering algorithms

For the clustering, we implemented a multi-algorithmic approach to ensure the robustness of the segments for each perspective. Primarily focusing on *K-Means* and *Agglomerative Hierarchical Clustering*. *K-Means* was selected for its efficiency in partitioning the large AIAI dataset. Since the number of clusters need to be set apriori, we created a graph of the *Elbow Curve* using the *Inertia Function*.

To validate these results, we used *Hierarchical Clustering*. Since this method has different methods of linkage, we plotted a graph comparing the results between *Ward*, *Complete*, *Average* and *Single* for each perspective. This method also groups similar data points into clusters, without needing to predefine the number of clusters, so we plotted a *Dendrogram* to determine the optimal number of clusters.

Finally, we implemented a *Self-Organizing Maps(SOM)*. Unlike the previous algorithms, *SOM* is a type of unsupervised artificial neural network. Its primary objective is dimensionality reduction: it transforms complex, high dimensional input data (our multiple customer attributes) into a discrete, low dimensional map. For that we need to calculate *SOM Quality Metrics: Quantization Error* and *Topographic Error*. The *Quantization Error* measures the average distance between data points and their assigned cluster centroids, serving as a metric for map resolution and data representation quality, while a *Topographic error* measures the proportion of data points for which the first and second *Best Matching Units (BMUs)* are not adjacent on the map grid.

4. Results & Validation

4.1. K-means

For the **demographic features**, the optimal number of clusters was determined to be three, based on the Elbow Curve analysis using the inertia function (**figure 1**). After applying the *K-means algorithm* with this configuration, the resulting clusters revealed clear demographic differentiation, supporting the validity and interpretability of the segmentation.

To further assess cluster quality, the *Euclidean distance* of each customer to the centroid of each demographic cluster was analyzed. For every observation, the smallest distance value identifies the nearest demographic centroid, and consequently determines the customer's cluster assignment. Additionally, repeated or closely valued distance patterns indicate subsets of customers sharing similar demographic profiles, reinforcing internal cluster consistency. Conversely, customers showing comparable distances to multiple centroids can be interpreted as borderline cases, whose demographic characteristics overlap across different segments.

Overall, this distance-based demographic validation confirms that the *K-means* solution captures meaningful demographic structure, albeit with lower separation than observed in the **behavioral**

perspective. These findings support the role of demographic variables as a complementary validation layer, enhancing the robustness, interpretability, and business relevance of the final segmentation when integrated with behavioral insights.

After analyzing the **demographic**, the next step was to look at data from a **behavioral perspective**. For this features, the optimal number of clusters was determined to be three too, based on the *Elbow Curve* analysis (**figure 2**). After applying the *K-Means algorithm* with this configuration, the resulting clusters provided a diagnostic understanding of the behavioral structures present in the data, confirming that meaningful segmentation patterns were achieved.

To further assess the quality of these **behavioral** clusters, the distance of each customer to the centroid of each cluster was analyzed. For every observation, the smallest distance value identifies the nearest **behavioral** centroid, which determines the customer's cluster assignment.

Overall, this distance-based **behavioral** validation confirms that the *K-means* solution captures a clear and distinct structure. These findings suggest that the **behavioral variables** offer a strong foundation for segmentation, providing the necessary differentiation to create actionable business segments.

4.2. Hierarchical Clustering

To explore the natural structure of the customer base, we applied *Hierarchical Clustering* to the **demographic features**, allowing us to build a hierarchy of similar groups without pre-specifying the number of clusters. The analysis began by defining the best linkage method to ensure accurate segmentation. We compared *Ward*, *Complete*, *Average*, and *Single* linkage methods using the R^2 metric to measure how well each explained the variance in the data. The results showed that *Ward linkage* consistently achieved the highest R^2 values across all cluster solutions, while *Single linkage* flatlined near zero, indicating a failure to separate clusters effectively (**figure 3**). Consequently, *Ward linkage* was selected for its ability to produce the most distinct segments.

Using *Ward linkage* and *Euclidean distance*, we examined the *Dendrogram* (**figure 5**) to determine the optimal number of clusters. The visual structure revealed a natural separation into 3 main clusters, evidenced by a relatively large increase in linkage distance after the third merge. This 3 cluster solution yields an R^2 of approximately **32.49%** (**table 1**), representing a moderate fit where distinct **demographic groups** exist, though significant individual variation ($\approx 67.5\%$) remains. This choice of $k=3$ aligns with previous findings, confirming the stability of these segments across different algorithms.

The final segmentation reveals three distinct customer profiles as we can see on **table 2**: **Cluster 0** : This segment exhibits the highest engagement in flight frequency (**TotalFlights: 0.60**) and rewards usage (**TotalPointsRedeemed: 0.37**) and has the longest tenure (**ProgramDaysDuration: 0.39**). However, their **Customer Lifetime Value (CLV)** is below average (-0.29), suggesting these are experienced travelers who fly often using cheaper tickets or points, limiting their profitability to the airline. Then, **Cluster 1**: This group is characterized by significantly below average values across all key metrics, including the lowest flight frequency (-1.13) and points redeemed (-0.69). Their defining trait is a very low **ProgramDuration (-0.76)**, indicating they are new customers who simply haven't had enough time to accumulate value or flights. Finally, **Cluster 2**: This is the most financially critical segment, defined almost exclusively by a massive **Customer Lifetime Value (1.75)**, which is nearly

two standard deviations above the mean. Despite having flight frequency slightly below average (-0.11), they generate immense value, likely by purchasing premium seats (e.g., Business Class) or high-value services rather than flying frequently on cheap fares.

On the other hand, for the **behavioral perspective**, the best linkage method after plotting the various hierarchical methods, was the *Ward*, as we can see in the **figure 4**. For the selection of the final number of clusters for the *hierarchical clustering* guided by the inspection of the *dendrogram* (**figure 6**) was 2.

The hierarchical clustering results demonstrate distinct behavioral differentiation across clusters as we can see in **table 2**, indicating meaningful segmentation driven by the **behavioral perspectives**. Customers grouped within the same cluster exhibit similar travel intensity and loyalty engagement patterns, while substantial dissimilarities are observed between clusters, reinforcing the internal consistency and external separation of the solution. Shorter linkage distances within clusters indicate strong behavioral similarity among customers, whereas larger distances between clusters confirm that the identified segments represent genuinely different **behavioral** profiles. This structure suggests that the clustering captures stable and interpretable behavioral patterns rather than noise-driven groupings.

Compared to demographic-based segmentation, the **behavioral** hierarchical clustering exhibits sharper boundaries and stronger separation (**figure 9**), reflecting the fact that travel behavior variables tend to be more discriminative and directly related to customer engagement with the airline. This enhanced separation supports the reliability of **behavioral features** as a primary driver of segmentation.

Overall, the *hierarchical clustering* applied to behavioral data yields well-defined and meaningful customer segments, validated through dendrogram structure and distance-based separation. These results provide a robust foundation for **behavioral** profiling and serve as a critical input to the final integrated segmentation framework, complementing **demographic** insights and supporting targeted marketing and loyalty strategies.

4.3. Self-Organizing maps

To complement the partition based and *hierarchical clustering* approaches, we implemented a *Self-Organizing map (SOM)*.

For the **Demographic features** on the *SOM quality metrics* (**table 4**) the most important result was the big drop on the *Topographic error (TE)*, which went from 0.9787 down to 0.2368 (a 75 % improvement after training). A high score here means the map is twisted and confusing. The new low score proves that the map is now smooth and organized correctly, so customers placed next to each other are actually similar. At the same time, the *Quantization Error (QE)* also got better, dropping from 0.7685 to 0.7154. This simply means that the model is now more precise and describes the customers more accurately. Since both numbers went down, we can trust that this model is stable and ready to be used for our analysis.

After, to visualize the individual variables we plotted a graph of the component planes (**figure 13**). For **Customer Lifetime Value** there was a specific red spot in the bottom-right corner, but that same spot was blue for the **TotalFlights**. This proves that there is a group of customers who pay a lot of money to the airline but actually take very few trips. Another interesting part of the **figure 13** is that the map for **ProgramDaysDuration** and **TotalFlights** are actually really similar, which means that staying with the airline is the main reason people fly. New customers(at the bottom of the map) are almost always inactive. On the other hand, for the **TotalPointsRedeemed**, while many people fly

often (the whole top area is red), only a small corner (top right) actually uses their rewards points. This shows that redeeming points is a special habit of only the most active users.

To visualize the structural boundaries within the *Self-Organizing Map*, we generated the *U-Matrix (Unified Distance Matrix)*(figure 15). While the previously analyzed Component Planes allow for the inspection of individual variables, the U-Matrix provides a holistic view of the geometric distances between the neurons themselves in the high dimensional feature space. The figure 15 reveals a clear topographic separation between the identified customer segments. The dominant feature is a large, continuous low distance basin (deep blue, values < 0.35) covering the left side of the map. This confirms that the "low Engagement" segment is highly uniform. Inactive customers tend to be inactive in very similar ways. A distinct vertical barrier of high distances (yellow/red, values > 0.65) bisects the map. This 'ridge' effectively acts as a structural wall, statistically isolating the active segments on the right from the inactive mass on the left. Finally, the presence of isolated high distance peaks (red hexagons) in the top left and distinct basins in the bottom right confirms that the first 2 groups previously mentioned are not merely extensions of the main population, but are distinct, well separated clusters, with unique statistical properties.

To see exactly how the customers are distributed across the different groups we plotted a *Hit Map* (figure 17). Analyzing this figure we can see that there is a highly concentrated group in the bottom right corner. We see one very dark hexagon containing 372 people. This is the highest number in a single spot. It tells us that this specific group of customers is almost identical to each other; they fit a very strict profile and do not vary much. The second largest population is found in the top section of the map which is full of dark green spots with high numbers between 150 and 200. Unlike the bottom right group, these customers spread different habits from one another. Finally, the map shows clear borders (light green) with numbers like 13 or 23 sitting between dark spots. These empty spaces act like walls, proving that customers usually fall clearly into one specific area, instead of being stuck in the middle.

Finally, for the **demographic features**, we plotted a *SOM K-Means clustering map* (figure 19). By comparing these colored zones with our previous analysis we can clearly identify the 3 clusters. The cluster 0 occupies the bottom-center of the map, with the color red, which represents the low engagement customers who have low flight activity. The cluster 1 occupies the bottom right section with the color yellow, which represents a specific niche we identified earlier. It captures the high value elite customers who generate high **Customer Lifetime Value**. Finally the cluster 2 that occupies the top left and center of the map in the color blue which aligns perfectly with the high activity areas for the **TotalFlights** and **ProgramDaysDuration**. This cluster contains the loyal frequent flyers. The fact that *K-Means* grouped the neurons into these three specific blocks confirms that the customer segments are topologically distinct. The map does not show a messy mix of customers. Instead, it shows clear, hard borders between the groups. This final validation proves that the 3 cluster solution is robust and geometrically stable.

For the **behaviour perspective**, the training quality (table 5) was evaluated using two metrics: *Quantization Error (QE)* and *Topographic Error (TE)*. The model showed significant improvement, with *QE* reducing from, approximately, 1.18 to 1.07 (a 9.4% improvement), indicating that the neurons successfully moved closer to the actual data points they represent. More notably, the *TE* dropped drastically from, approximately, 0.97 to 0.19 (an 80.7% improvement), meaning the map successfully organized itself so that similar points (customers) are now placed next to each other.

Then, visual analysis of the resulting map provided deep insights into customer behaviors (figure 14). The heatmaps revealed a strong correlation between **Income** and **TotalPointsRedeemed** in the lower left region, as well as distinct seasonality where **Flights_Winter** and **Flights_Fall** shared nearly

identical high density patterns distinct from other seasons. The U-Matrix (**figure 16**) confirmed the presence of clear boundaries between these identified groups, indicating robust cluster separation.

The *Hit Map* (**figure 18**), however, identified a significant distribution, where most of the map was evenly populated, one specific outlier cluster contained over 1400 samples, representing a highly concentrated group of customers. Finally, a *K-Means algorithm* (**figure 20**) was applied to the trained *SOM* to partition the map into three distinct personas. The “*Purple*” group is the big crowd where everyone is alike. The “*Yellow*” group acts as a middle ground, and the “*Red*” group on the edges shows customers with different habits. This process successfully simplified the high dimensional data into a clear strategic map for customer segmentation.

4.4 Merging Clusters analysis

After looking at customers in two different ways (per perspectives: demographic and behaviour), we needed to bring them together. Having separate groups for **demographic** and **behavior** can be confusing to manage, so the final goal was to merge them into a single, easy to use, list of customer types.

To combine these **3 behavior groups** with our **4 demographic groups**, we created a table to see where they overlapped. We found some strong connections, for example, one specific **behavior** group was almost entirely made up of people from just one **demographic** group. To merge them properly, we calculated the average traits for every possible combination and used a method called *Hierarchical Clustering*. This method grouped the most similar combinations together and drew a *dendrogram* to guide our decision. The chart revealed that the best approach was to organize everyone into 4 final merged clusters.

To visualize our merged clusters we plotted the **figure 10** and **figure 21**, and we can take some insides:

- Cluster 0: **Solo Travelers**. This segment displays a high **SoloFlightRatio** (indicated by the deep green), paired with low seasonal flight activity (purple). Despite their high **demographic** and **behavioral** cluster scores, they are not high-volume fliers but are very distinct in their solo travel preference.
- Cluster 1: **Low-Engagement Users**. Characterized by neutral or low values across almost all **behavioral features**, this group represents the "baseline" customer. They show moderate duration in the program but minimal flight activity across all seasons.
- Cluster 2: **Consistent Loyalists**. This is a highly active segment showing strong positive engagement (green) across all seasons (Fall, Spring, Summer, and Winter). They are consistent fliers with a high **TotalFlights** count and high **TotalFlightsWithCompanions**, suggesting they are likely leisure travelers or families.
- Cluster 3: **High-Value Potential**. This group shows moderate engagement across most features. While they don't peak in seasonal flights like Cluster 2, they maintain steady participation and contribute significantly to the overall customer base.

5. Strategic Recommendations

Strategic recommendations have been developed for each of AIA's customer segments based on the clustering analysis. These strategies focus on personalized marketing, service customization, and loyalty program optimization.

For the **Solo travelers**, the niche group that travels alone (highest **SoloFlightRatio**) but also shows the lowest seasonal participation and point redemption, we decided to implement "Solitude & Speed" campaigns. Focus on the ease of travel for one, emphasizing lounge access and fast-track security as primary benefits. Use targeted emails with destination-specific "hidden gems" suitable for solo explorers. Offer priority boarding and seat selection that favors single-aisle convenience or quiet zones. Provide mobile-first check-in experiences to minimize friction during travel. We also decided that it would be interesting to create "Solo Milestones " since they redeem points less frequently, offer lower redemption thresholds for high-utility single-person rewards, such as extra baggage allowance or single-entry lounge passes, to encourage program engagement.

For the **Low-Engagement Users** who have high demographic scores but are largely inactive across all metrics, including seasonal flights and point redemption. We decided to focus on "Re-activation and Aspiration" for the marketing approach to use demographic data to send personalized "Where to Next?" promotions based on their profile (e.g., location or education level). Introduce "First-Flight" discount vouchers to break the inactivity cycle. We also thought it would be interesting to implement a "Tier Protection" or "Status Kickstart" promotion where they can earn double points for a limited time to jumpstart their engagement. Simplify the redemption process for non-flight rewards (e.g., retail partners) to show immediate program value.

For the **Consistent Loyalists**, the largest and most valuable segment, showing high positive activity across all metrics, including seasonal flights and companion travel for marketing, we thought to use a "Retention and Community" messaging to acknowledge their loyalty with "Thank You" campaigns. Encourage their habit of traveling with others by offering "Companion Multiplier" points during peak seasons. It would be interesting to have dedicated customer service lines and high-touch "surprise and delight" perks (e.g., complimentary onboard beverages or priority upgrades). Focus on family/group travel conveniences, such as grouped check-in. We could also introduce an exclusive "Platinum" or invitation-only tier above the current Aurora status. Offer benefits that scale with flight frequency, such as complimentary companion vouchers or transferable points for family members.

And finally for **high-Value Potential** which is a small elite segment with the highest demographic and behavioral scores. They have high potential for premium services despite currently flying less than Consistent Loyalists. In the marketing we thought that we should focus on Market First Class and Business Class upgrades directly to this group. Use "Invite-only" events or early access to new route launches to appeal to their high-value profile. We could also offer high-end concierge services, private lounge access, and chauffeur transfers to and from the airport. Ensure a seamless, premium end-to-end travel experience. We could offer a "Status Matching" or immediate elevation to the highest tier based on their profile. Provide bespoke rewards such as private jet partnerships or luxury hotel stays to align with their high-value potential.

6. Conclusion

We have successfully transformed a complex and fragmented customer database into a clear, actionable strategy. By validating **behavioral** patterns through the **Self-Organizing Map** and merging them with **demographic** profiles, we condensed the entire customer base into **four** distinct strategic personas. This transition from theoretical analysis to a unified merged Cluster model provides the business with a precise roadmap. It allows for the immediate replacement of generic marketing with targeted campaigns, ensuring that resources are allocated efficiently based on the specific value and habits of each of the four identified groups.

7. Annexes:

7.1. Annex Statements

7.1.1. AI Usage Statement

- AI tools were used for coding syntax of more complex blocks of code, code debugging assistance and to brainstorm ideas on how to best visualize the data(ChatGPT), and report proofreading and improvement (Grammarly and ChatGPT).

All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking.

7.1.2. Contribution Statement

Ana Rita Dias (Student ID: 20250480) & Catarina Santos (Student ID: 20250501)

We made all the steps of the project together. All members contributed to collaborative discussions and ideation as well as for the development and review of the final report.

7.1.3. Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.

7.2 Annex: Dataset Information

Feature	Description
Loyalty#	Unique customer identifier for loyalty program members
First Name	Customer's first name
Last Name	Customer's last name
Customer Name	Customer's full name (concatenated)
Province or State	Customer's province or state
City	Customer's city of residence
Latitude	Geographic latitude coordinate of customer location
Longitude	Geographic longitude coordinate of customer location
Postal code	Customer's postal/ZIP code
Gender	Customer's gender
Education	Customer's highest education level (Bachelor, College, etc.)
Location Code	Urban / Suburban / Rural classification of customer

	residence
Income	Customer's annual income
Marital Status	Customer's marital status (Married, Single, Divorced)
LoyaltyStatus	Current tier status in loyalty program (Star > Nova > Aurora)
EnrollmentDateOpening	Date when customer joined the loyalty program
CancellationDate	Date when customer left the program
Customer Lifetime Value	Total calculated monetary value of customer relationship
EnrollmentType	Method of joining the loyalty program
TotalFlights	Total number of flights taken over the observed period
TotalFlightsWithCompanions	Total number of flights taken with companions
TotalSoloFlights	Total number of flights taken solo
TotalDistanceKM	Total distance traveled in kilometers
TotalPointsAccumulated	Total loyalty points earned
TotalPointsRedeemed	Total loyalty points spent
TotalDollarCostPointsRedeemed	Total dollar value of points redeemed
AvgFlightsPerMonth	Average number of flights per month
AvgPointsPerMonth	Average points earned per month
MaxFlightsInMonth	Maximum number of flights in a single month
MaxPointsInMonth	Maximum points earned in a single month
CompanionFlightRatio	Proportion of flights taken with companions relative to total flights
SoloflightRatio	Proportion of flights taken solo relative to total flights
PointsRatio	Proportion of points redeemed to points accumulated
PointsRedeemedPerFlight	Average points redeemed per flight
AvgCompanionsPerFlight	Average number of companion flights per flight
Flights_Fall	Total number of flights during fall
Flights_Spring	Total number of flights during spring
Flights_Summer	Total number of flights during summer
Flights_Winter	Total number of flights during winter

7.3. Figures

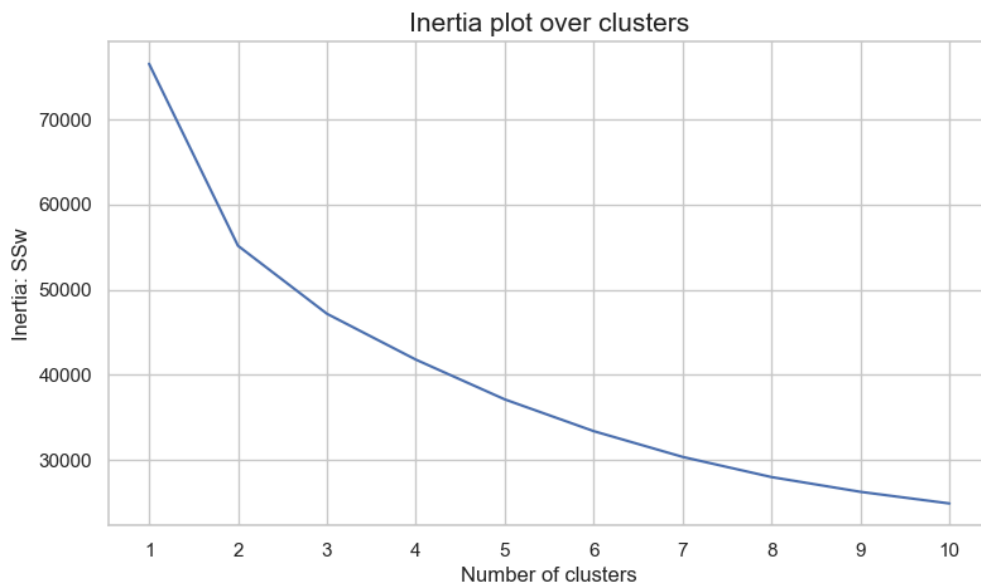


Figure 1: Inertia Plot for the **Demographic** perspective.



Figure 2: Inertia Plot for the **Behavior** perspective.

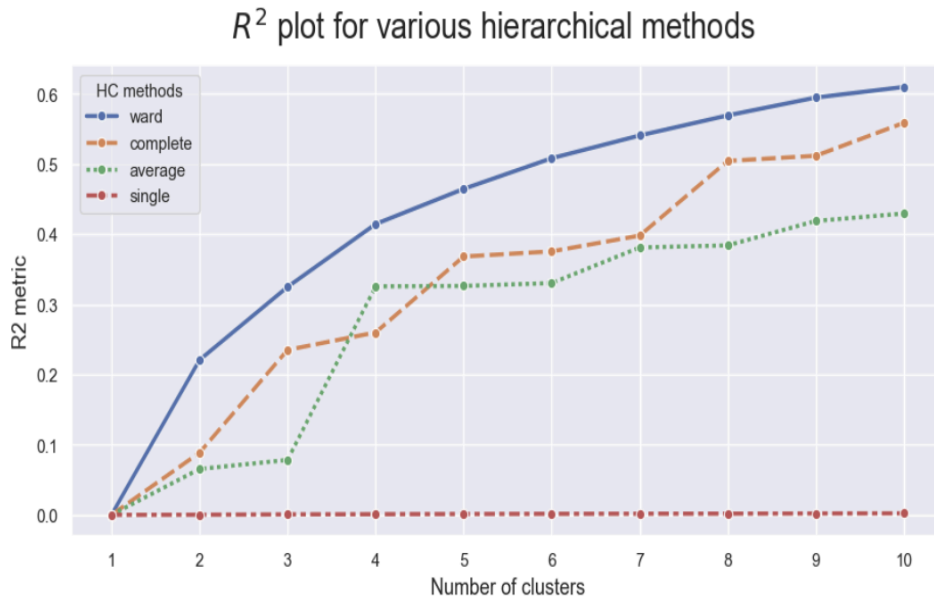


Figure 3: R^2 plot for various hierarchical methods for the **Demographic perspectives**.

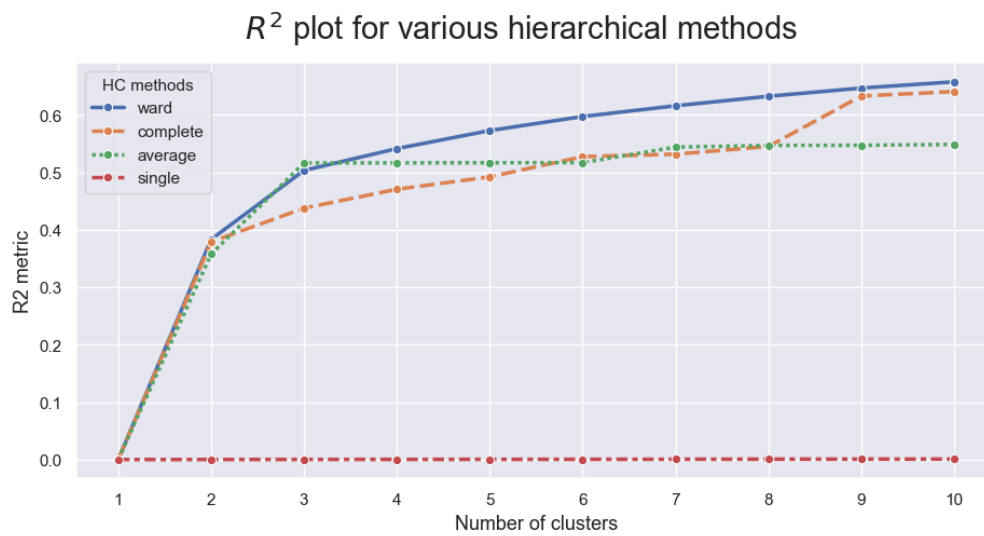


Figure 4: R^2 plot for various hierarchical methods for the **Behaviour perspectives**.

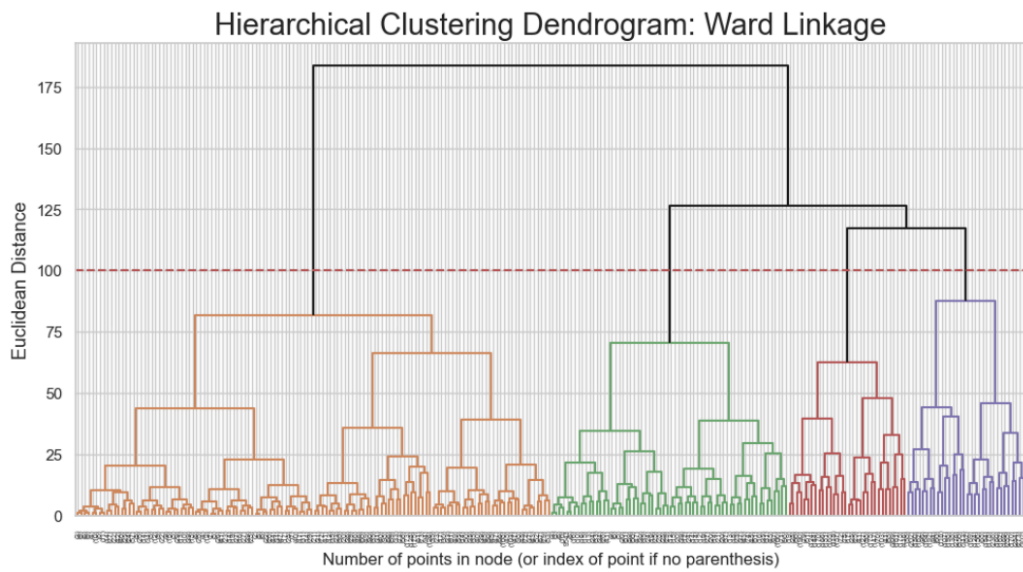


Figure 5: Hierarchical Clustering Dendrogram for the **Demographic Perspectives**.

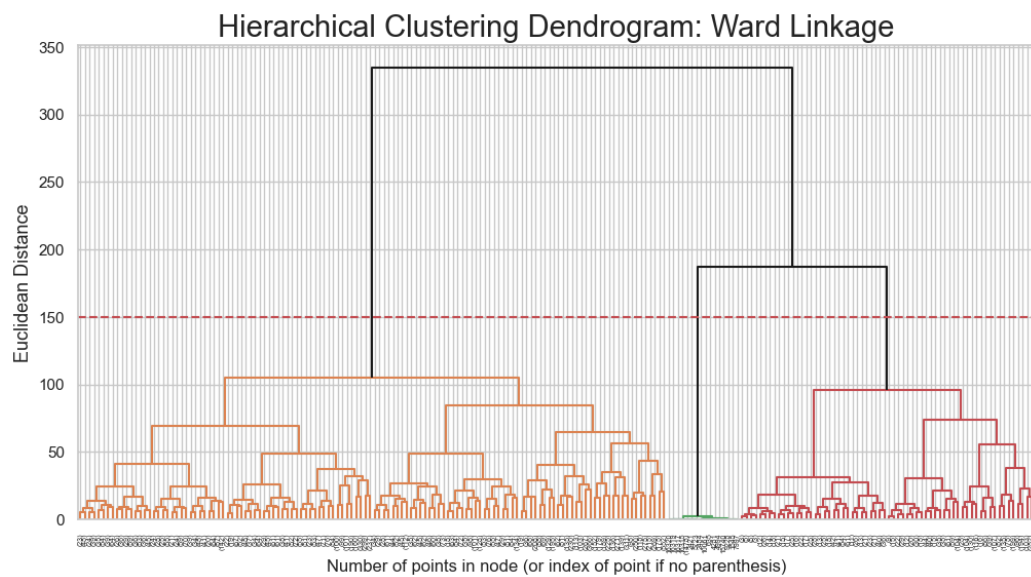


Figure 6: Hierarchical Clustering Dendrogram for the **Behaviour Perspectives**.

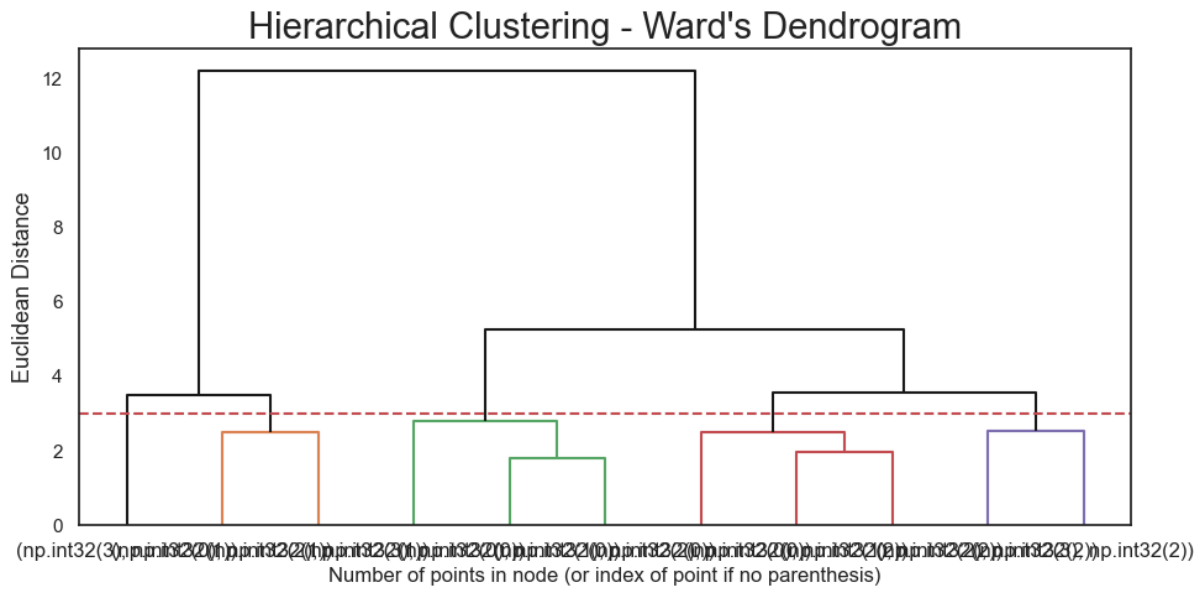


Figure 7: Hierarchical Clustering Dendrogram for the **merged**

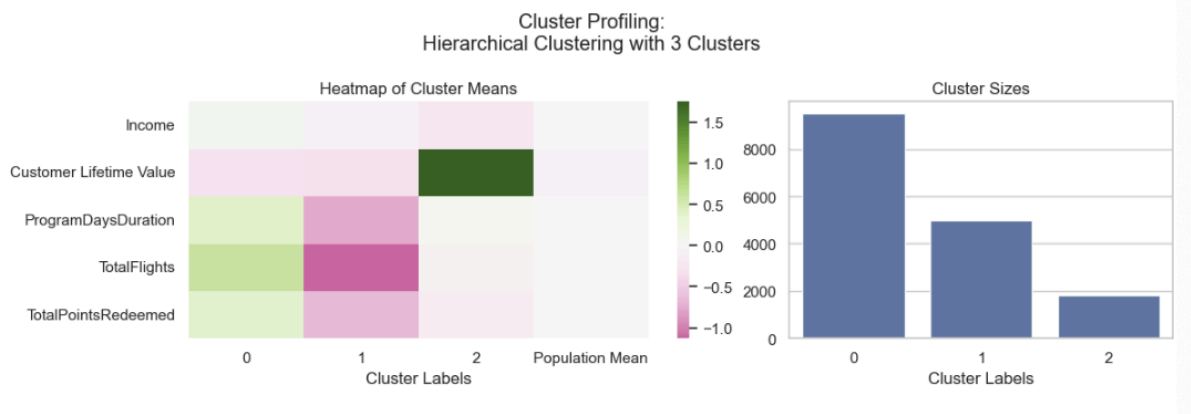


Figure 8: Hierarchical Cluster Profiling for the **Demographic Perspectives**.

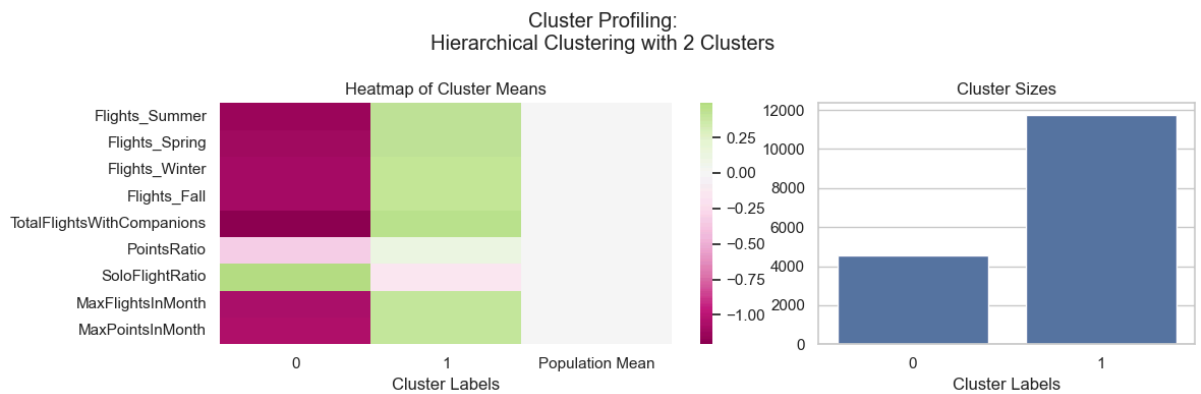


Figure 9: Hierarchical Cluster Profiling for the **Behaviour Perspectives**.

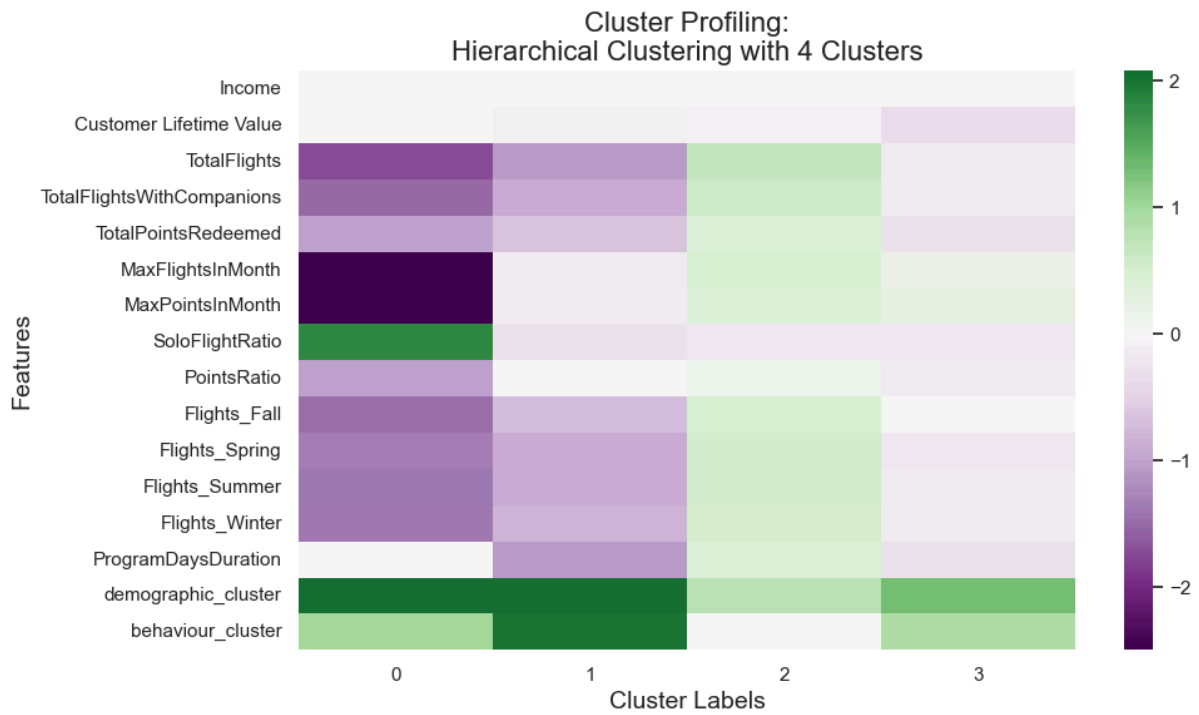


Figure 10: Hierarchical Cluster Profiling for the merged.

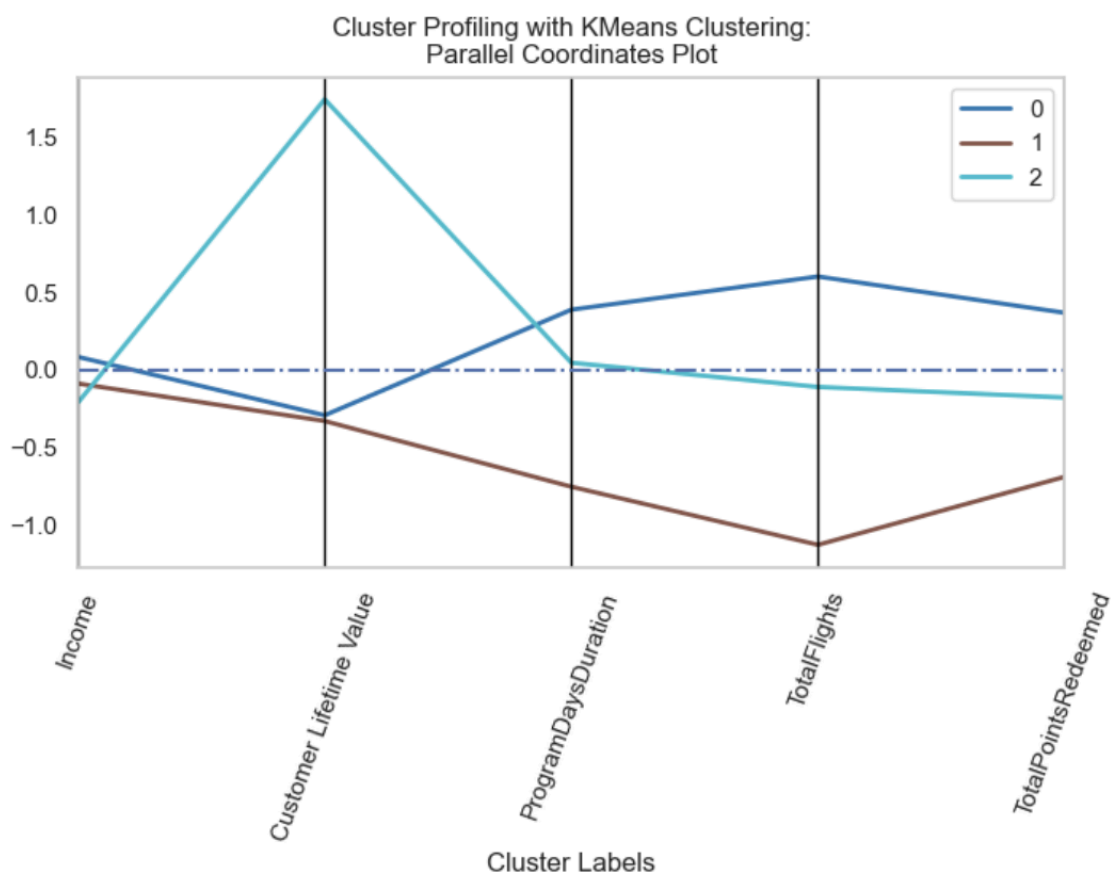


Figure 11: Cluster Profiling with K-means for Demographic Segments.

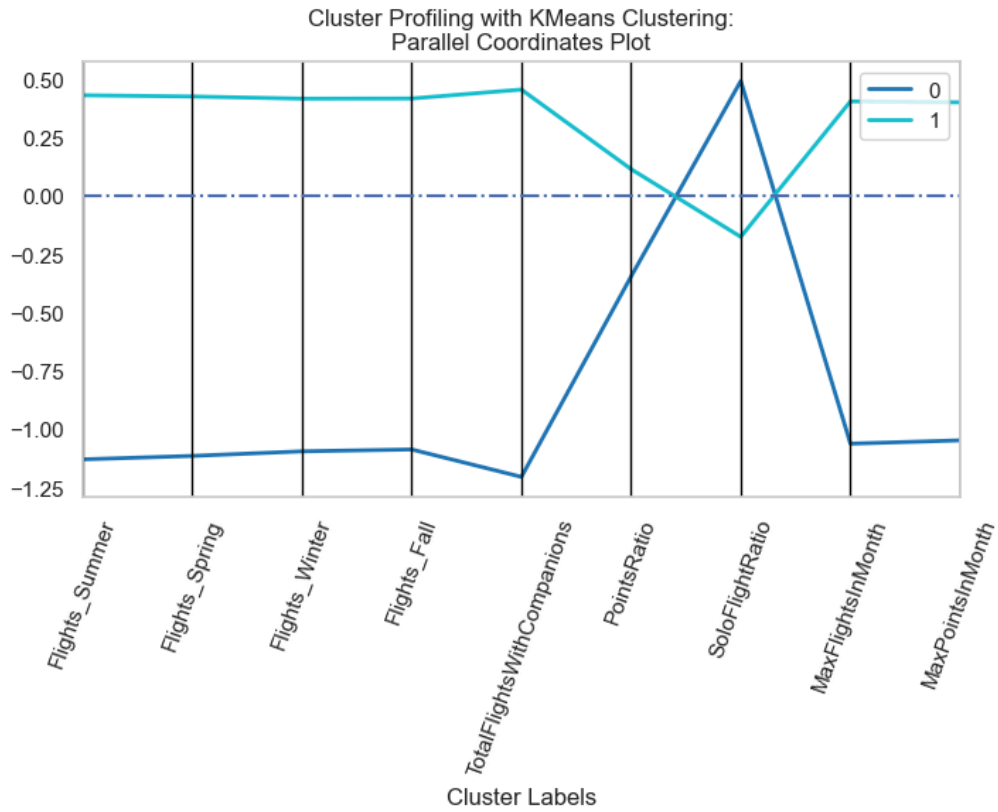


Figure 12: Cluster Profiling with K-means for **Behaviour Segments**.

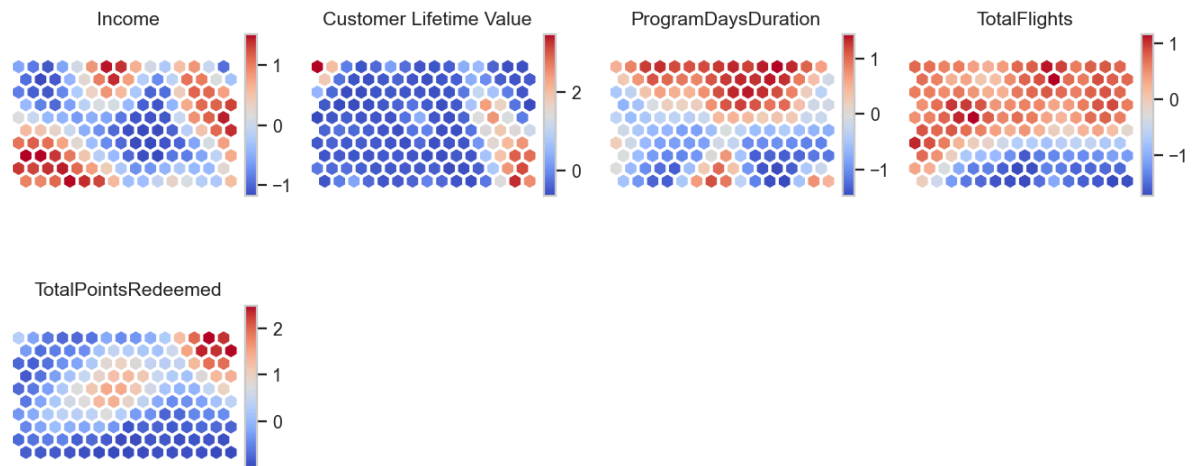


Figure 13: Component planes of individual variables for the **demographic features**.

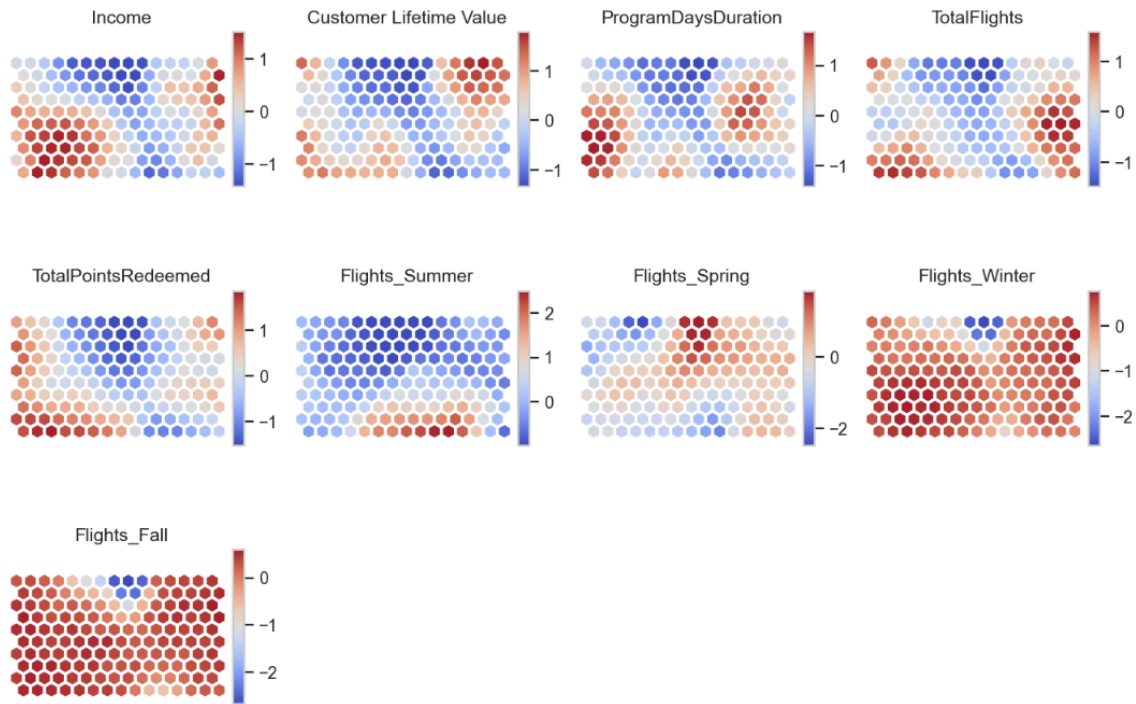


Figure 14: Component planes of individual variables for the **behaviour** features.

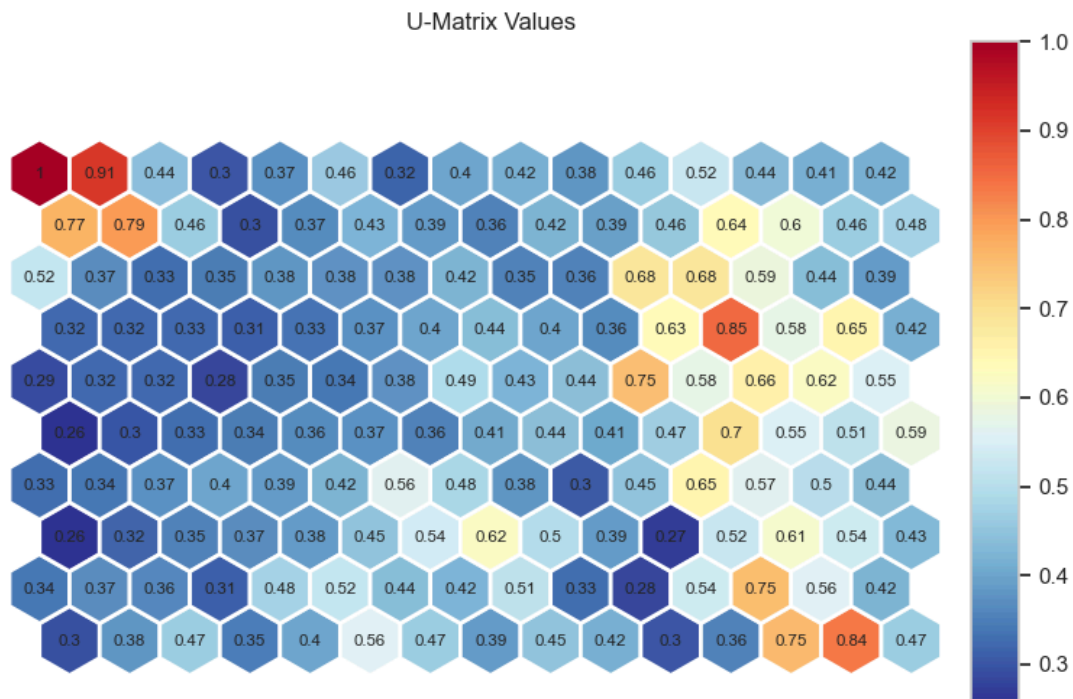


Figure 15: U-matrix visualization for the **demographic** features.

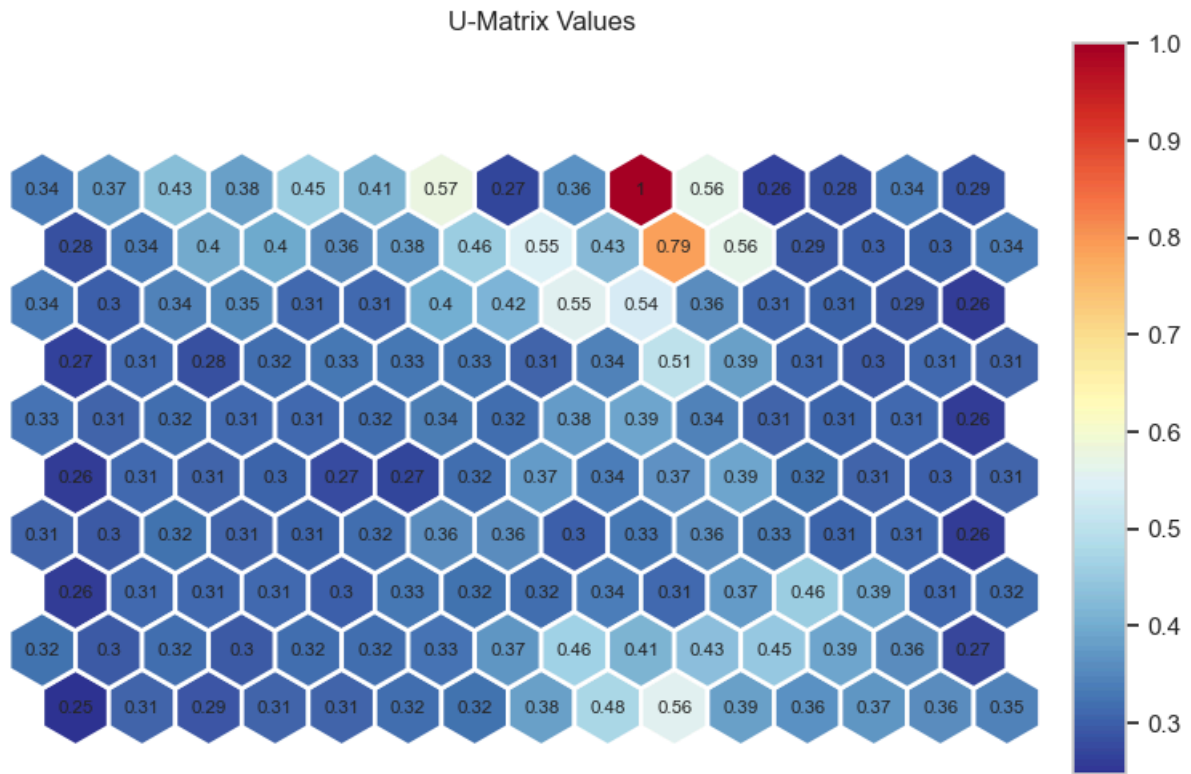


Figure 16: U-matrix visualization for the **behaviour features**.

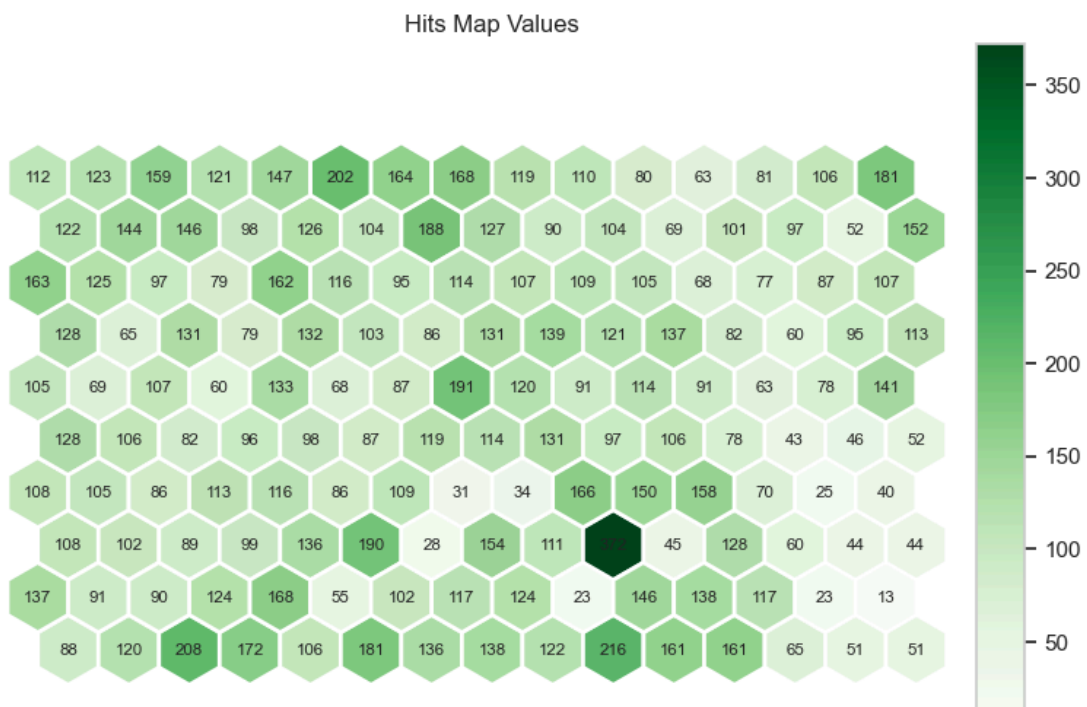


Figure 17: Hit-map for the **demographic features**.

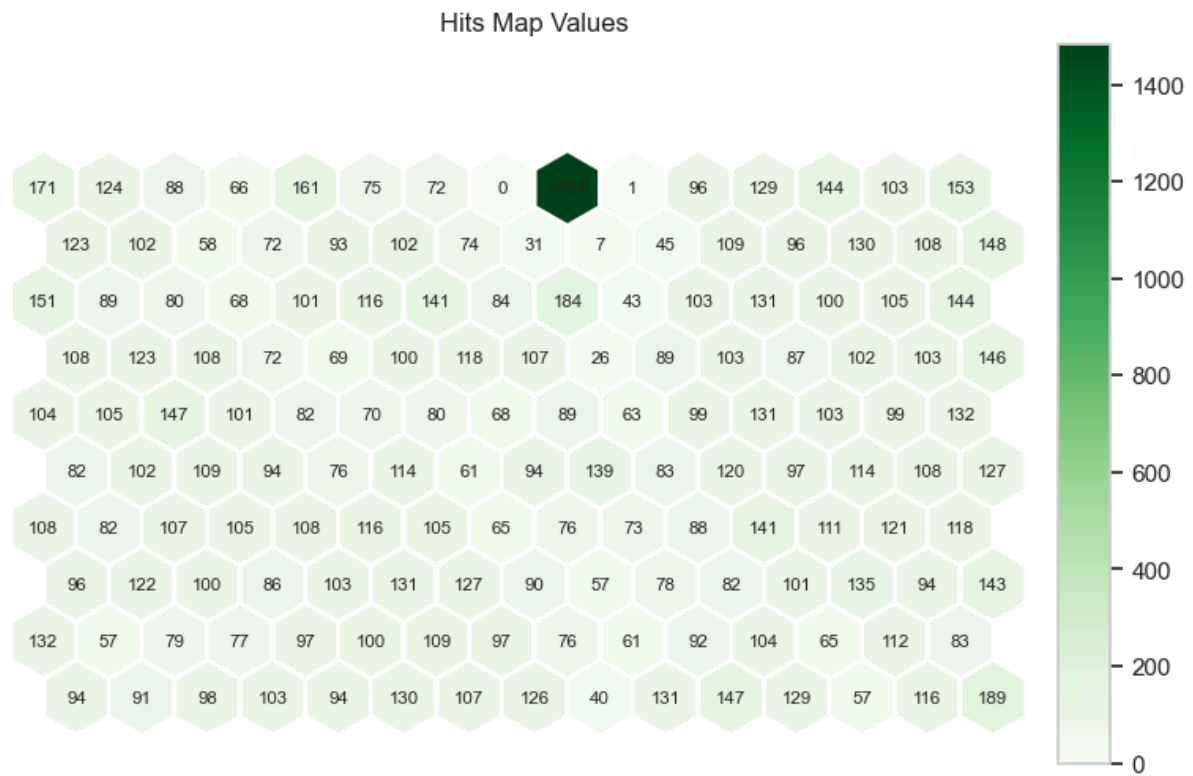


Figure 18: Hit-map for the **behaviour** features.

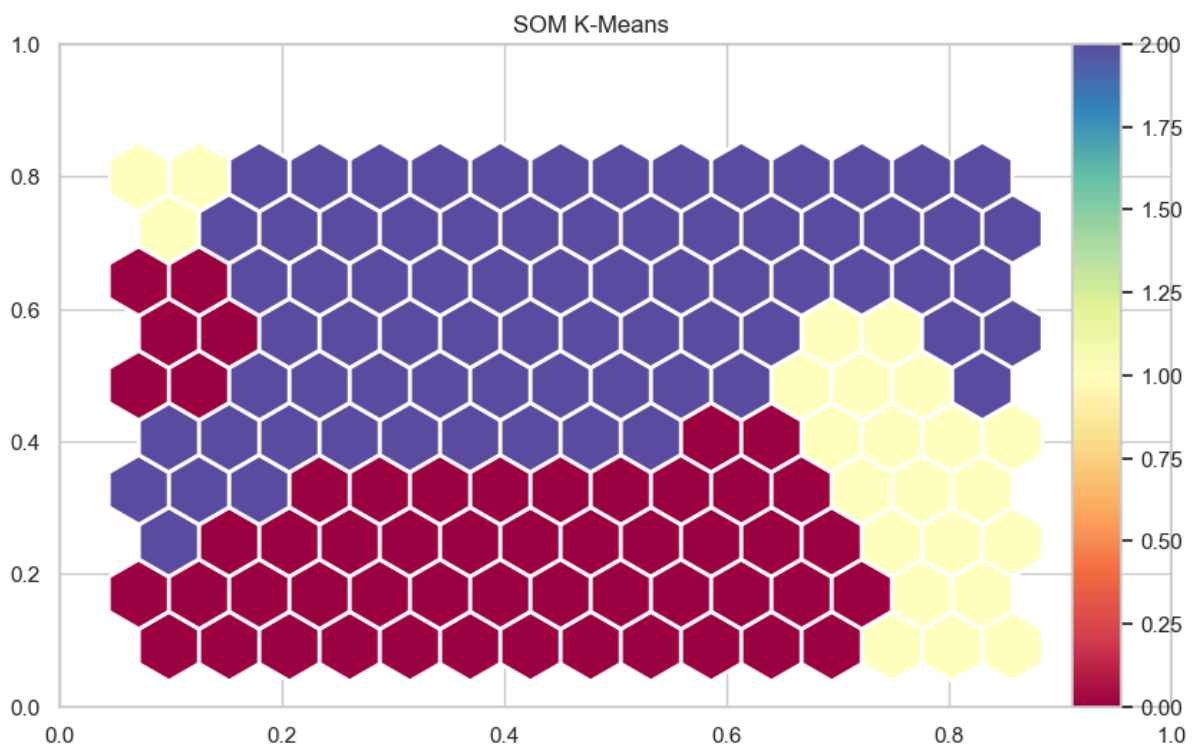


Figure 19: SOM k-means for the **demographic** features.

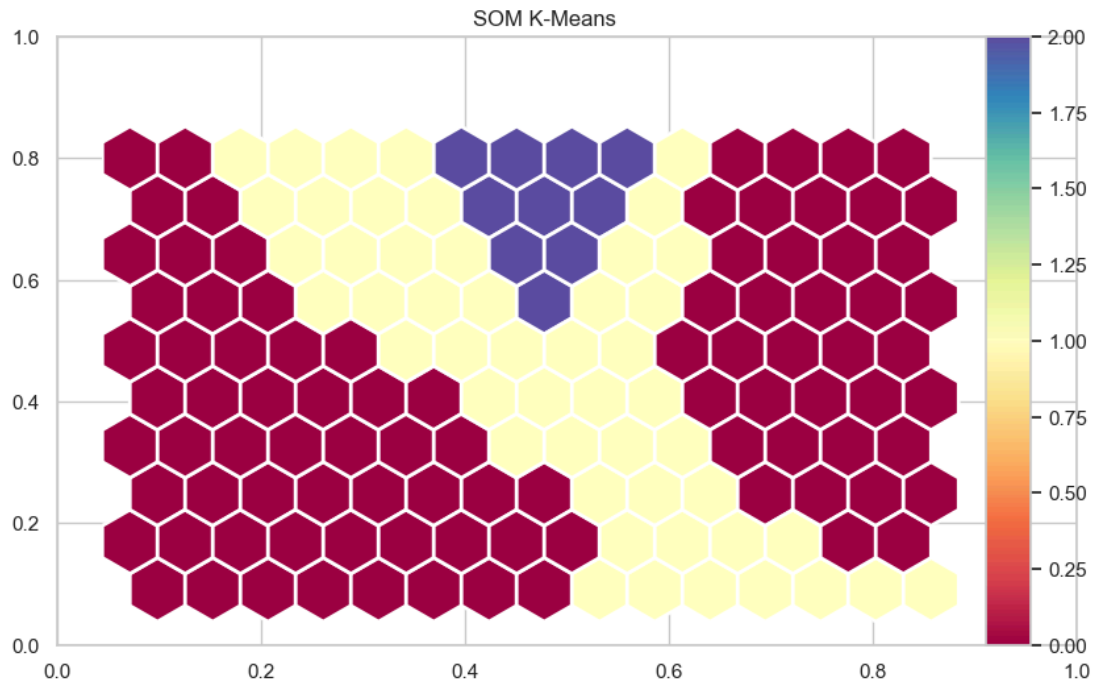


Figure 20: SOM k-means for the behaviour features.

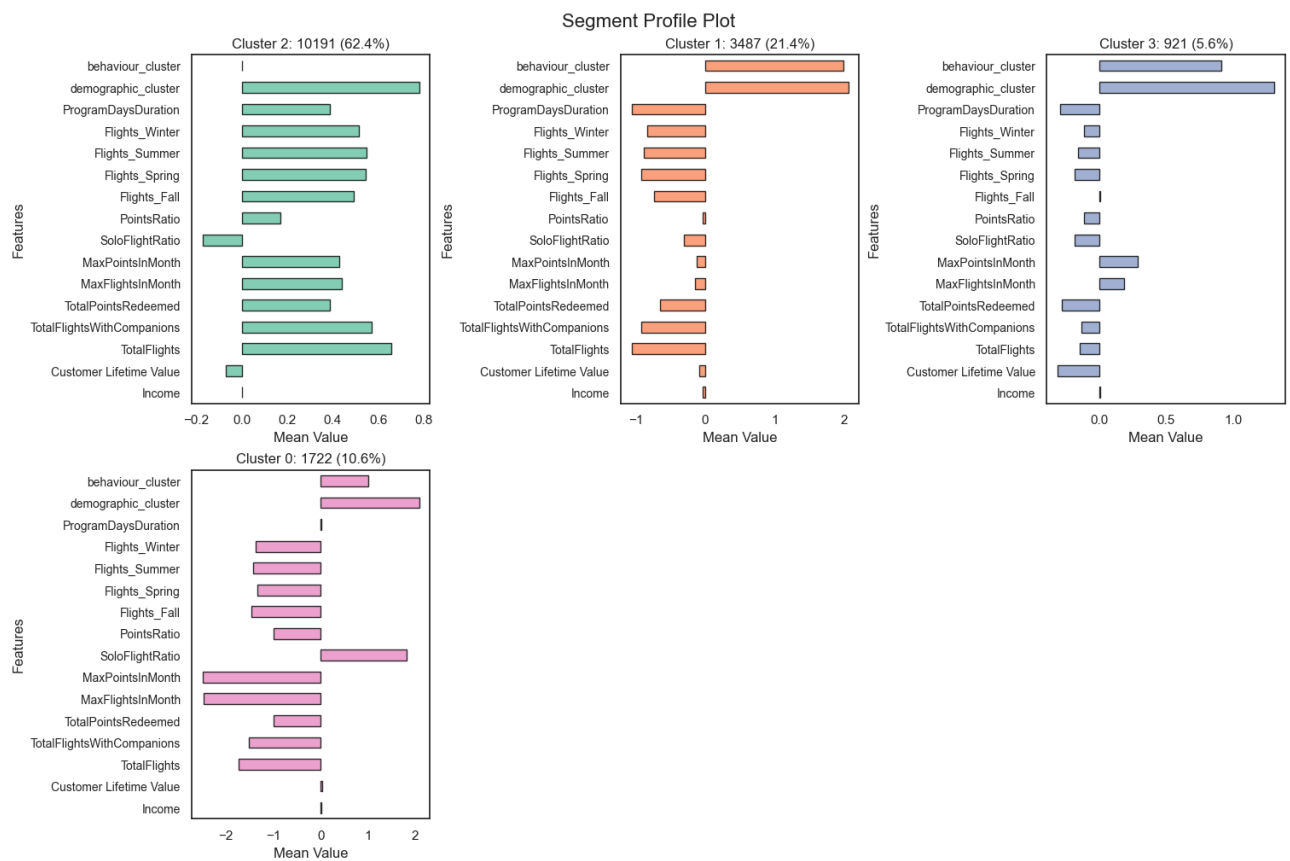


Figure 21: Segment Profile plot for merged.

7.4. Tables

	ward	complete	average	single
1	0.00000	0.00000	0.00000	0.00000
2	0.22078	0.08857	0.06538	0.00019
3	0.32491	0.23511	0.07828	0.00073
4	0.41454	0.26002	0.32564	0.00095
5	0.46462	0.36825	0.32632	0.00124
6	0.50835	0.37557	0.33036	0.00148
7	0.54079	0.39808	0.38116	0.00162
8	0.56942	0.50449	0.38413	0.00176
9	0.59484	0.51184	0.41893	0.00201
10	0.60993	0.55879	0.42950	0.00219

Table 1: Comparison of likeaged methods for **Demographic** Segments.

labels	0	1	2
Income	0.08307	-0.08875	-0.21111
Customer Lifetime Value	-0.29372	-0.33091	1.74611
ProgramDaysDuration	0.38801	-0.75504	0.04547
TotalFlights	0.60214	-1.12994	-0.11086
TotalPointsRedeemed	0.36789	-0.69102	-0.17932

Table 2: Comparison of Mean Feature Values across **Demographic** Segments.

labels	0	1
Flights_Summer	-1.13131	0.43066
Flights_Spring	-1.11648	0.42544
Flights_Winter	-1.09683	0.41607
Flights_Fall	-1.08881	0.41677
TotalFlightsWithCompanions	-1.20640	0.45513
PointsRatio	-0.34681	0.11381
SoloFlightRatio	0.49171	-0.17673
MaxFlightsInMonth	-1.06402	0.40448
MaxPointsInMonth	-1.04974	0.40033

Table 3: Comparison of Mean Feature Values across **Behavioral** Segments.

	Before training	After training	Improvement (%)
Quantization error	0.7685	0.7154	6.9%
Topographic error	0.9787	0.2368	75.8%

Table 4: SOM metrics for the **demographic** features.

	Before training	After training	Improvement (%)
Quantization error	1.1792	1.0688	9.4%
Topographic error	0.9683	0.1869	80.7%

Table 5: SOM metrics for the **behaviour features**.

behaviour_cluster	0	1	2
demographic_cluster			
0	3989	12	216
1	5333	0	204
2	501	1547	3264
3	869	163	223

Table 6: cross-tabulation table.